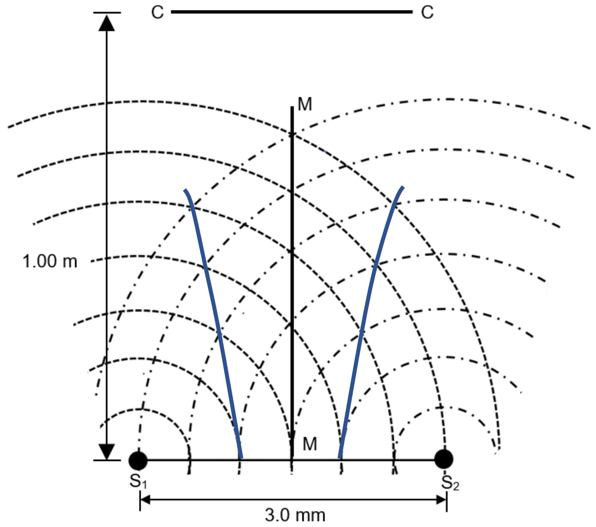
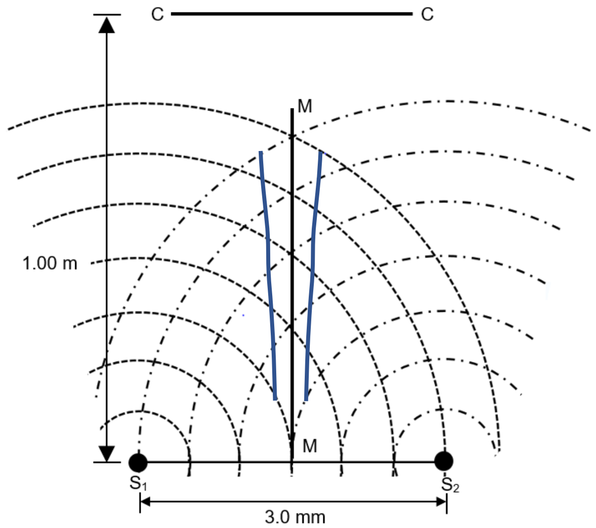
Question	Answer	Marks
8(a)(i)	When two or more waves overlap (meet) The resultant displacement at any point and instance is the vector sum of the displacements caused by the individual waves at that point at that instance.	B1 A1

8(a)(ii)	The two waves must be of the same type (i.e both waves must be electromagnetic/sound waves) The two waves must be coherent. The two waves must have similar amplitudes. If the two waves are transverse waves, they must either be unpolarised or polarized in the same plane. (1 mark for each correct condition. Maximum of 3 marks)	**B3**
8(b)(i)	180° (The two sources are in antiphase.) (As the waves that arrive at line MM have a path difference of 0, since a minima is detected, the sources must be in antiphase).	**A1**
8(b)(ii)1.	 The lines above are possible EE lines. (Drawn line should be at least 2 wavelengths long).	**A1**
8(b)(ii)2.	 The lines above are possible FF lines. (Drawn line should be at least 2 wavelengths long.)	**A1**
8(b)(iii)	Line FF: Antinodal line	**B1**
8(b)(iv)	A stationary wave/ standing wave. (As the interference pattern is produced by two coherent sources of waves meeting along a line).	**A1**
8(b)(v)	Number of wavefronts between S_1 and S_2 is 6. Distance between S_1 and S_2 = 6 λ = 3.0 mm λ = 3.0 / 6 = 0.5 mm (Shown)	**M1**

8(b)(vi)	As a stationary wave is formed, the distance between each minima = ½ λ = 0.25 mm Number of half-wavelengths in 3.0 mm = 3.0/0.25 = 12 Number of minimas detected (not inclusive of the sources) = 11 (Draw out a diagram if you are uncertain. If the sources are counted, there will be a total of 13 minimas)	**M1** **M1** **A1**
8(b)(vii)	(As the distance of the sources to line CC is much greater than the distance between both sources, the equation $\Delta y = \frac{\lambda L}{a}$ suitable) Using equation $\Delta y = \frac{\lambda L}{a}$ $\Delta y = \frac{(0.5\times10^{-3})(1.0)}{3.0\times10^{-3}}$ $\Delta y = 0.167$ m	 **M1** **A1**
8b(viii)1.	For CC: The distance between maximas (or minimas) will reduce. (Based on $\Delta y = \frac{\lambda L}{a}$, as *a* reduces, Δy increases.) For S_1S_2: There is no change to the distance between maximas. (The interference pattern is that of a stationary wave, distance between antinodes is half a wavelength).	 **A1** **A1**
8b(viii)2.	For CC: The contrast between the maximas and minimas will be reduced. (Due to incomplete cancellation at minima and the reduced amplitude at maximas). For S_1S_2: The contrast between the maximas and minimas will be reduced. (Due to incomplete cancellation at minima and the reduced amplitude at maximas).	 **A1** **A1**
	Max Marks	**20**

Setter: Yen Ling Marker:		
Qn 9	**Answer**	**Marks**
9(a)		